

Polar and Rectangular Forms of Equations

Convert each equation from polar to rectangular form.

1) $\tan \theta = 2$

$$y = 2x$$

2) $r = 4\cos \theta - 4\sin \theta$

$$(x - 2)^2 + (y + 2)^2 = 8$$

3) $r = -2\cos \theta$

$$(x + 1)^2 + y^2 = 1$$

4) $r = 2\cos \theta + 2\sin \theta$

$$(x - 1)^2 + (y - 1)^2 = 2$$

Convert each equation from rectangular to polar form.

5) $(x - 1)^2 + (y + 1)^2 = 2$

$$r = 2\cos \theta - 2\sin \theta$$

6) $x = y^2$

$$r = \cot \theta \csc \theta$$

7) $x = y^2$

$$r = \cot \theta \csc \theta$$

8) $y = \frac{x^2}{5}$

$$r = 5\tan \theta \sec \theta$$

Convert each equation from polar to rectangular form.

9) $r = 4\csc \left(\theta + \frac{\pi}{6} \right)$

$$y = -\frac{x\sqrt{3}}{3} + \frac{8\sqrt{3}}{3}$$

10) $r = 2\sin \left(\theta + \frac{\pi}{4} \right)$

$$\left(x - \frac{\sqrt{2}}{2} \right)^2 + \left(y - \frac{\sqrt{2}}{2} \right)^2 = 1$$

11) $r^2 = 5\sec (2\theta)$

$$x^2 - y^2 = 5$$

12) $r^2 = 4\sec (2\theta)$

$$x^2 - y^2 = 4$$